

ROB 545: Kinematics, Dynamics, & Control

Statics

Joe Davidson

Goals for today

- Discuss other important uses of the Jacobian matrix
 - Static analysis
 - Singularities
 - Manipulability

Review of the Jacobian matrix

- Given the configuration θ of the robot, the 6-vector $J_i(\theta)$, which is column i of $J(\theta)$, is the twist $\mathcal{V}$ when $\dot{\theta}_i = 1$ and all other joint velocities are zero
- The twist $\mathcal{V}$ is found the same way as the joint screw axes were determined in forward kinematics, except the screw axes of the Jacobian depend on the joint variables θ (for F.K. $\theta = 0$)
- This is known as the geometric Jacobian

Review of the Jacobian matrix

The i^{th} column of $J_s(\theta) \in \mathbb{R}^{6 \times n}$ is given by

$$J_{si}(\theta) = Ad_{e^{[S_1]\theta_1} \dots e^{[S_{i-1}]\theta_{i-1}}}(S_i) \text{ for } i = 2, \dots, n \text{ with the first column } J_{s1} = S_1$$

The i^{th} column of $J_b(\theta) \in \mathbb{R}^{6 \times n}$ is given by

$$J_{bi}(\theta) = Ad_{e^{-[B_n]\theta_n} \dots e^{-[B_{i+1}]\theta_{i+1}}}(B_i) \text{ for } i = n - 1, \dots, 1 \text{ with } J_{bn} = B_n$$

The body and space Jacobians are related via

$$J_s(\theta) = [Ad_{T_{sb}}]J_b(\theta)$$

$$J_b(\theta) = [Ad_{T_{bs}}]J_s(\theta)$$

Using the Adjoint representation

In our rigid body lectures, we saw that $[V_s] = T[V_b]T^{-1}$, which can be manipulated into

$$\begin{bmatrix} \omega_s \\ v_s \end{bmatrix} = \underbrace{\begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix}}_{\text{Adjoint representation of } T} \begin{bmatrix} \omega_b \\ v_b \end{bmatrix}$$

Adjoint representation of T

Let $T_1, T_2 \in SE(3)$ and $V = (\omega, v)$. Then $Ad_{T_1} \left(Ad_{T_2}(V) \right) = Ad_{T_1 T_2}(V)$

The Jacobian tells us a lot about the robot!

Consider a situation where an external force is applied to the end-effector. What are the joint torques required to resist this external force?

Using the *conservation of power* (power = force × velocity):

$$\text{power at the joints} = (\text{power to move robot}) + (\text{power at the end-effector})$$

We merge the moment and force acting on a rigid body into a six-dimensional **spatial force**, or **wrench**:

$$F = \begin{bmatrix} m \\ f \end{bmatrix} \in \mathbb{R}^6$$

Static analysis

At static equilibrium, no power is being used to move the robot (excluding torque required to offset gravity)

So, power at joints = power at end-effector

$$\tau^T \dot{\theta} = F_b^T \mathcal{V}_b$$

Substituting $\mathcal{V}_b = \mathbf{J}_b(\theta)\dot{\theta}$ yields:

$$\tau^T \dot{\theta} = F_b^T \mathbf{J}_b(\theta) \dot{\theta}$$

$$\tau = \mathbf{J}_b^T(\theta) F_b$$

$$\tau = \mathbf{J}_*^T(\theta) F_*$$

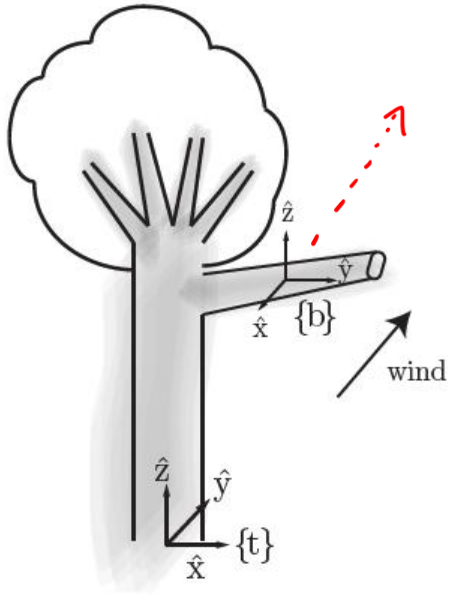
* = b is the Jacobian and wrench represented in $\{b\}$

* = s is the Jacobian and wrench represented in $\{s\}$

Must be true for all $\dot{\theta}$

$$(\tau^T)^T = (F_b^T \mathbf{J}_b(\theta))^T$$
$$\tau = \mathbf{J}_b^T(\theta) F_b$$

Example: Calculating wrenches



Consider a tree {t} with a frame {b} attached to its branch. The figure shows a strong wind that applies a force of 100 N at the center of frame {b}. Assume the branch has a mass of 50 kg centered at frame {b} as well. The position of the branch frame {b} in the tree frame {t} is $p_{tb} = (2, 1, 3)$ m.

(a) What is the wrench F_b due to the wind and the branch's weight?

Let's say $g = 10 \text{ m/sec}^2$

The branch weight is: $(50 \text{ kg})(10 \text{ m/sec}^2) = 500 \text{ N}$

The wind force is: 100 N

These forces are applied at the center of {b}

$$m_b = [0 \ 0 \ 0]^T$$

$$F_b = [-100 \ 0 \ -500]^T \text{ (N)}$$

$$F_b = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -100 \\ 0 \\ -500 \end{bmatrix}$$